

Handling low-coverage uncertainty in methylation estimates, Part 2

Ezra Parsons — 28/06/2026. All results on simulated data. Code & figures:

<https://github.com/EzraParsons1/rockfish-lowcov-methylation>

Abstract

Grouping reads across co-methylated CpG sites is the most promising way to fight low-coverage noise in an epigenetic clock, but it requires determining which sites are co-methylated. A discovery method (correlate sites across specimens then cluster the correlated ones), when scored against known blocks of co-methylated sites, is able to determine blocks reliably ($F1 > 0.9$) once the gap between block and background correlation exceeds ~ 0.5 . This threshold is invariant to sequencing depth and average block size. It shifts only with the number of specimens (better accuracy with more specimens). The gap is measurable in real data and must be tested before it is certain that grouping sites will improve the clock. The two other methods tested to reduce low-coverage noise, correcting mean reversion and replacing elastic-net with a likelihood-based clock, failed to beat a simple elastic-net regression. As in report 1, magnitudes are simulator specific. However, methods and relative performance are transferable.

The problem: the clock reverts to the mean at low coverage

At low coverage the elastic-net clock systematically pulls predictions toward the population mean (young fish read too old, old fish too young) and the effect steepens as depth drops.

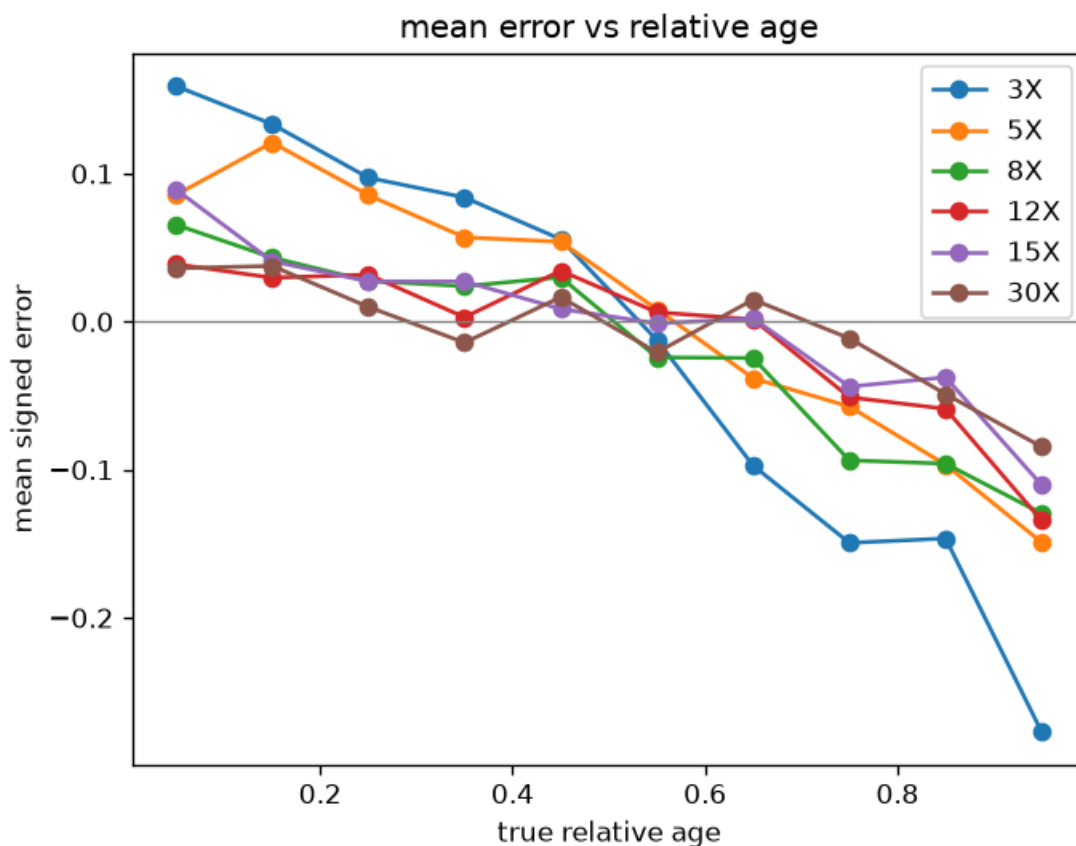


Figure 1: Mean signed error vs true age, at several depths. The young-positive to old-negative tilt is regression to the mean, growing as coverage falls. The low-coverage accuracy problem is shrinkage, not random scatter: can it simply be undone?

Can the low-coverage bias be calibrated out?

I used a least-error algorithm to determine the best slope to be applied to the clock's age predictions. The intention was for the algorithm to select a slope which corrected the mean-reversion bias generated by the elastic-net regression. It always selected a slope of ~ 1 . Meaning the best way to correct the bias is not to correct it. This is because the mean reversion bias is intentionally added to reduce variance caused by low-coverage.

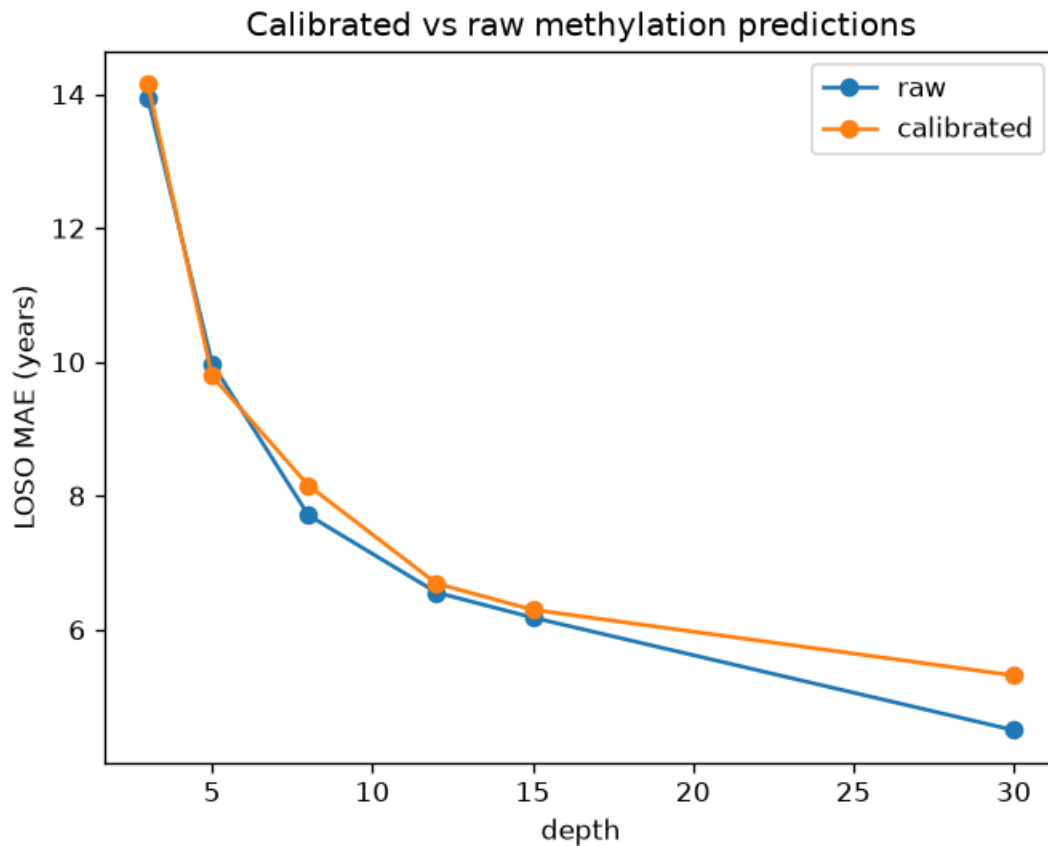


Figure 2: Effect of post-hoc calibration on the clock, by depth. The calibration slope stays ~ 1 at every depth and MAE is flat-to-worse.

Does a likelihood model beat the elastic net?

I built a binomial-MLE clock which fits a per-site reference line (methylation vs relative age) on the training species then, for each test fish, scans candidate ages and picks the one maximizing the summed binomial log-likelihood of its reads. This method never forms a point estimate of methylation probability at a site and weighs sites by coverage automatically. It extracts roughly the same age signal as the elastic net but does not regress to the mean. Again, this causes it to have worse results: shrinkage is the right response to noise, so the MLE has higher variance and performs worse in absolute error.

Coverage	Model	r (age signal)	slope	Error std	MAE in years
3X	EN	0.77	0.55	0.18	13.3
3X	MLE	0.78	1.17	0.27	19.7
30X	EN	0.94	0.89	0.10	3.9
30X	MLE	0.93	1.14	0.14	9.2

(The MLE is better on short-lived species, where un-shrunk predictions help in absolute years. It worked in single cells due to their extreme low depth where elastic-net breaks.)

Both this and the calibration result point the same way: the error is intrinsic to the noise, so the only thing left is to reduce noise at the source. (*Table from binomial_MLE.py.*)

Which is better: breadth or depth?

Given that, where should limited sequencing be spent: more sites (breadth) or more reads per site (depth)? I swept a 10,000-site genome over depth (3–30X) and fraction of sites observed (0.1–1.0), scoring a leave-one-species-out clock (v3b shrinkage, 5 seeds).

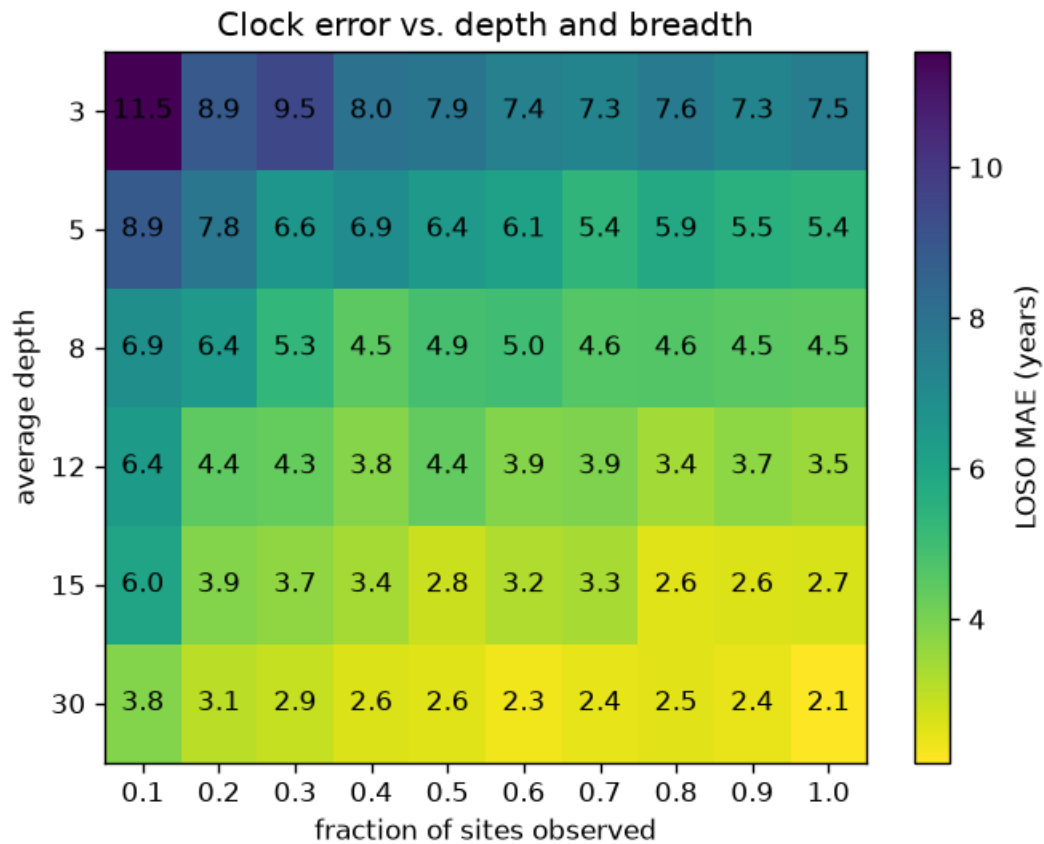


Figure 3: Clock error over fraction-of-sites-observed × depth. Depth beats breadth, especially past ~0.4 of sites observed. per unit of sequencing it's better to read fewer sites deeper (half the sites at 5X vs all sites at 5X costs only ~1 year). grouping reads across co-methylated sites converts breadth into effective depth, therefore, it is the most valuable method for reducing low-coverage bias. The only question is whether or not they can be grouped effectively.

Finding co-methylation blocks

To find blocks of correlation, I wrote an algorithm to compute the site-by-site methylation correlation across individuals, connect sites whose correlation exceeds a threshold (~0.5 is best, unrelated to the required correlation contrast), then take connected components as discovered blocks. Because this is simulated, it can score discovery directly against the true blocks (pair-based precision/recall/F1).

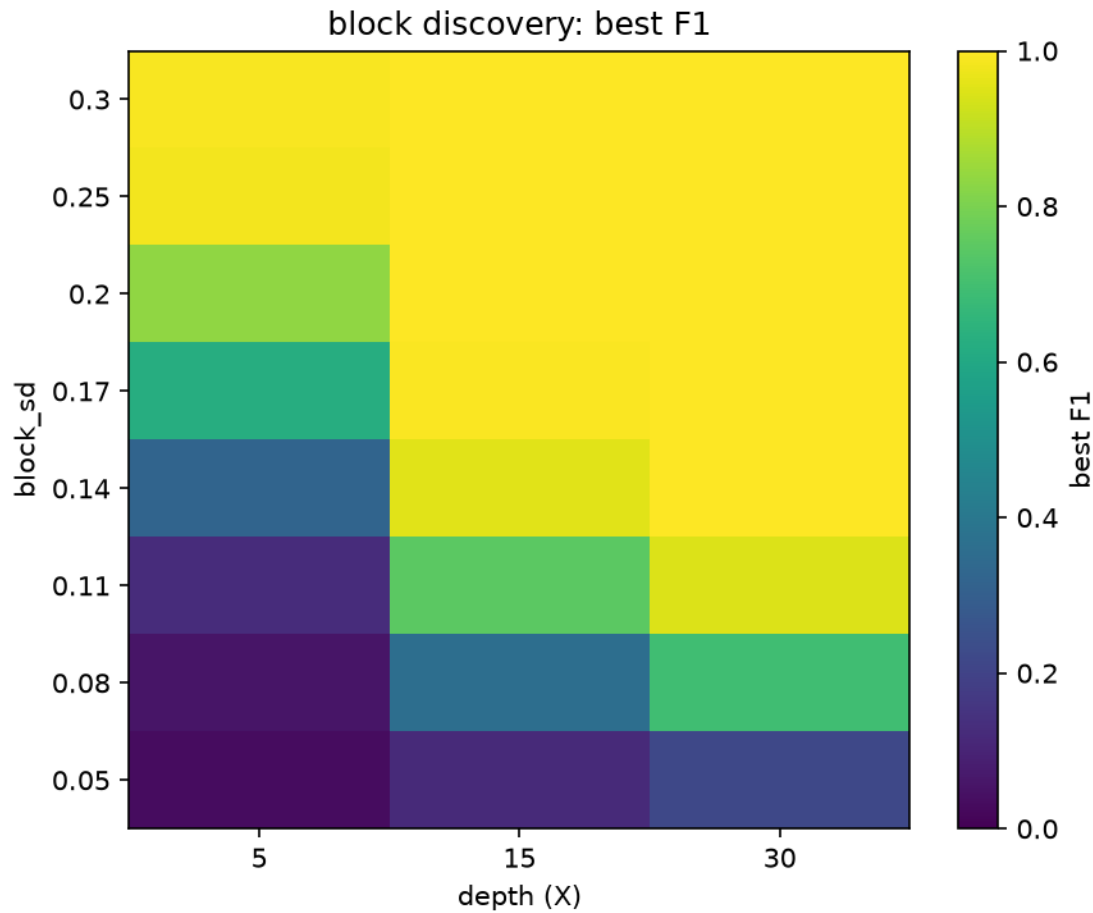


Figure 4: Best F1 over block correlation strength $\times$ depth. Discovery is near-perfect at strong correlation and high depth and falls off a sharp cliff below a certain threshold. However, the cliff's location depends on depth, so it isn't a portable rule on its own.

To create a transferrable threshold, it needs to express a value which doesn't change with depth. Within-block correlation drifts with depth because background correlation grows with depth alongside block correlation. The better measure is contrast between within and across block correlations.

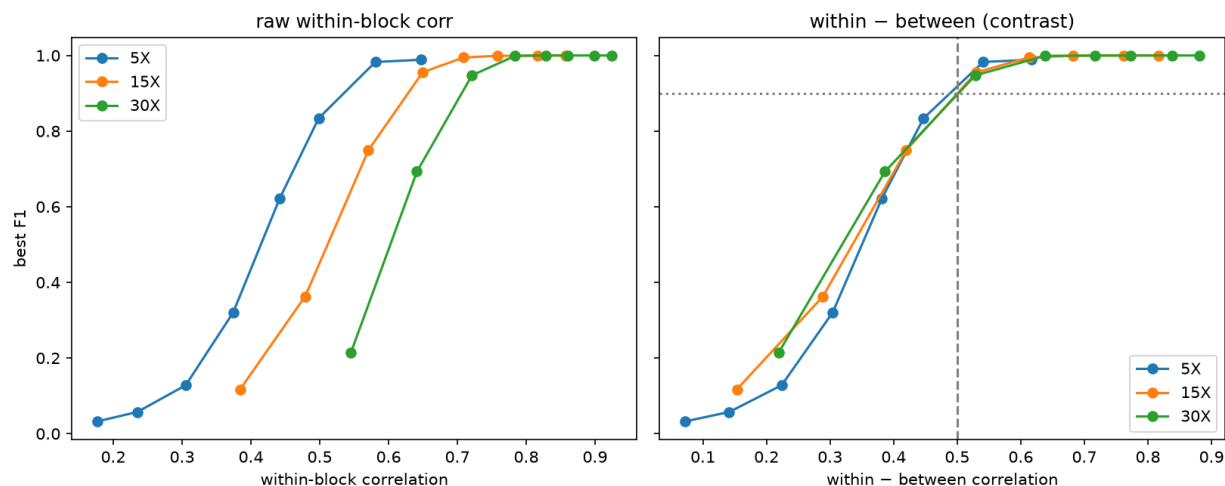


Figure 5: F1 vs raw within-block correlation (left) and F1 vs contrast (right), colored by depth. Against raw within-block correlation the depth curves fan apart; against contrast they collapse into one curve, crossing F1 = 0.9 at ≈ 0.5 , the same at 5X / 15X / 30X.

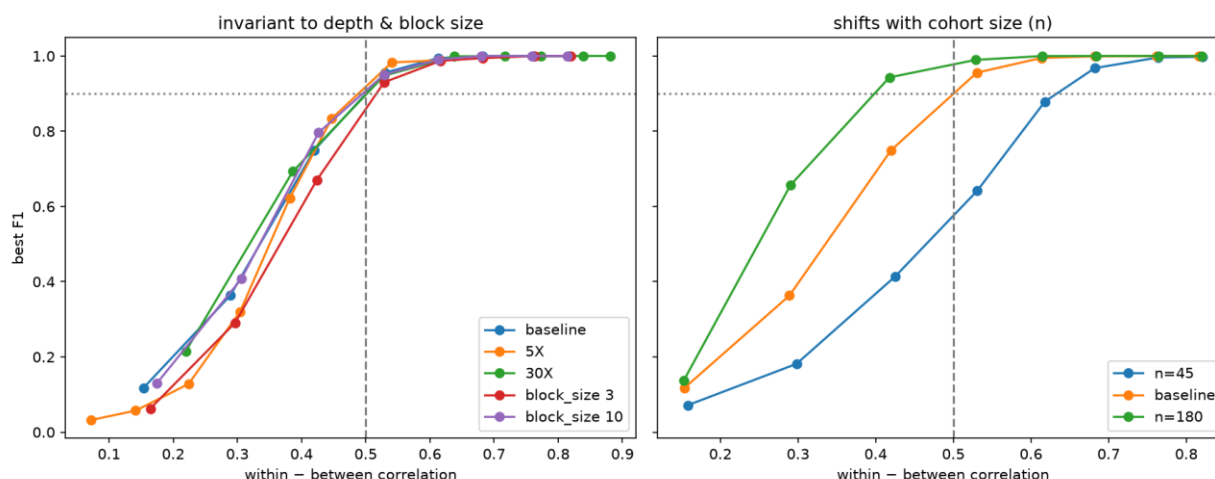


Figure 6: The collapse holds beyond depth. F1 vs contrast is also invariant to block size (left); the only factor that shifts it is cohort size (right). Fewer fish make the correlation estimates noisier ($\sim 1/\sqrt{n}$), so a larger gap is needed: ≈ 0.68 at 45 specimens, ≈ 0.5 at 90, ≈ 0.42 at 180. (Depth attenuates the mean correlation, which the contrast absorbs; cohort size changes the spread of the estimates, which contrast cannot absorb. Therefore, depth collapses but cohort size shifts.)

Recommendation

- To determine whether or not grouping sites can benefit the clock, measure the within-region vs between-region CpG correlation gap in the real data. If it is greater

than ~ 0.5 at 90 fish (~ 0.68 if we have ~ 45 fish), block discovery should recover co-methylation structure.

- Don't fight the low-coverage bias at the output or use a likelihood clock.

Caveats and next steps

- The ~ 0.5 threshold uses the ground-truth-optimal discovery threshold, so it's a ceiling on what is technically possible.
- Whether grouping discovered blocks improves the clock depends on whether co-methylation is age correlated. Grouping raises effective depth but only helps if co-methylated sites share an age signal, a property of the real data. Any other sites are already removed by the elastic-net regression.
- The next step is to determine the true threshold and effectiveness of grouping on real data.

Reproduce

Figures: `error_vs_age.py` (Fig 1), `calibration.py` (Fig 2), `binomial_MLE.py` (MLE table), `breadth_depth.py` + `heatmap.py` (Fig 3), `collapse.py` (Figs 4–5), `collapse_robust.py` (Fig 6). Modules: `simulate.py`, `estimators.py`, `clock.py`, `binomial_MLE.py`, `grouping.py`.